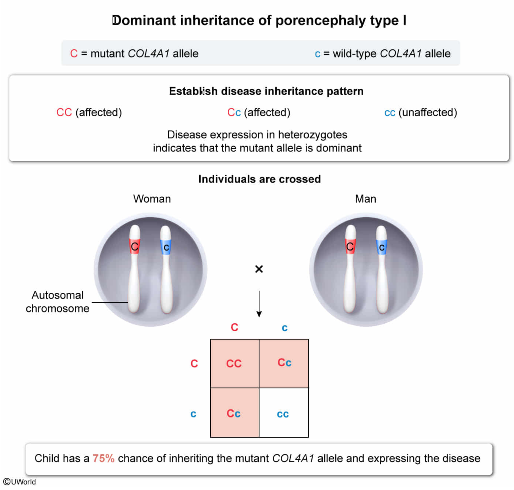
Porencephaly type I, a disease caused by a mutation in the autosomal *COL4A1* gene, is characterized by the development of noncancerous, fluid-filled growths in brain tissue. A woman expressing this disease is heterozygous for the *COL4A1* mutation. If she has a child with a man who is also heterozygous for the mutation, what is the probability that the child will express the disease?

- A. 25% [17%]
- B. 50% [9%]
- ✔ C. 75% [69%]
- D. 100% [3%]

Correct
69% answered correctly

Explanation:



DNA mutations can lead to the production of new alleles (ie, versions) of a gene. Alleles, which may be dominant or recessive, can be found on autosomes, sex chromosomes, or mitochondrial DNA.

The question describes porencephaly type I, a disease caused by a mutation in the autosomal gene *COL4A1*. In this scenario, a **woman who expresses this disease** is heterozygous for the *COL4A1* mutation (ie, carries one wild-type allele [c] and one mutant allele [C]). Therefore, the *COL4A1* mutation that causes this disease must be **dominant** because heterozygotes (Cc) would *not* express recessive mutations (ie, recessive mutations would be *masked*).

A heterozygous parent (Cc) has a **50%** chance of passing on the mutant allele (C). Because this mutation is dominant, the child must inherit only **one copy** of the **mutant** allele to express the disease. Therefore, to express this disease, the child could inherit either a *single COL4A1* mutation (ie, one C allele from the heterozygous mother *or* father) or *two* copies of the mutation (ie, one C allele from each heterozygous parent). If this heterozygous woman (Cc) has a child with a man who is also heterozygous (Cc) for this mutation, the probability that the child will express the disease can be calculated as follows:

$$\mathrm{P(child\ expresses\ disease) = P(inherits\ single\ mutant\ C\ allele) + P(inherits\ two\ mutant\ C\ alleles)}$$

Accordingly, each component probability can be calculated separately:

$$\mathrm{P(child\ inherits\ single\ mutant\ C\ allele) = P(inherits\ C\ from\ mother\ and\ c\ from\ father) + P(inherits\ C\ from\ fathe}$$

$$= (0.5 \times 0.5) + (0.5 \times 0.5) = 0.25 + 0.25 = 0.50$$

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$$P(\text{child inherits two mutant C alleles}) = P(\text{inherits C allele from both mother and father})$$
$$= 0.5 \times 0.5 = 0.25$$

Next, the sum of the component probabilities can be calculated as follows:

$$P(\text{inherits single mutant C allele}) + P(\text{inherits two mutant C alleles}) = 0.50 + 0.25 = 0.75$$

Therefore, the child has a **75%** chance of **expressing the disease**.

(Choices A, B, and D) These options result from *incorrect* calculations.

Educational objective:
In autosomal dominant inheritance, an individual must inherit only one copy of the dominant allele to express the trait. In addition, there is a 50% chance that offspring will inherit the mutation from a heterozygous parent.

Time spent:0 sec
Subject:
Biology

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